MBDP metagenomics course

2026

Course information

- Practicals on Mon | Wed | Fri 9.00–16.00
- Self-learning materials for the days in between
- Instructions and materials can be found from:

https://github.com/MBDP-bioinformatics-courses/MBDP_Metagenomics_2026

HUOM!/OBS!/ACHTUNG!/NOTE!
Read the instructions carefully before running anything

Learning outcomes

By completing this course, you will:

- Have a basic understanding of metagenomic sequencing technologies and bioinformatic approaches to analyse metagenomic data
- Be able to plan and execute a metagenomic sequencing project depending on the research questions.
- Have an up-to-date knowledge on the bioinformatic tools and best practices for the analysis of metagenomes.
- Be able to choose and critically evaluate new tools and approaches for specific research question
- Have confidence to learn and implement new bioinformatic methods using available documentation

Introductions

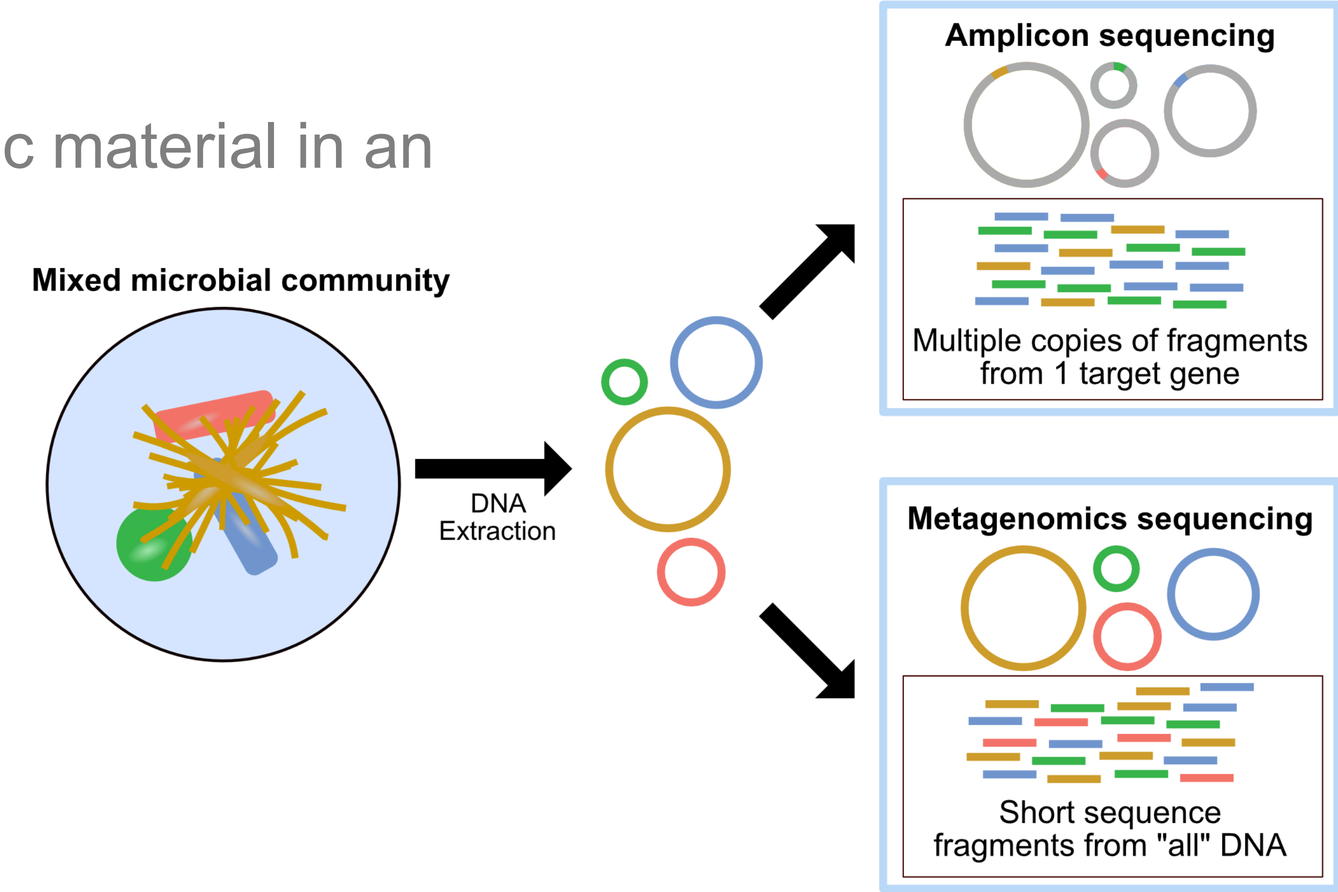
- Introduce yourself briefly (<1 min)
  - Name and affiliation
  - What is your PhD about?
  - OPTIONAL: Expectations for the course?

Metagenomics

Short introduction

Metagenomics

- Study of all genetic material in an environment

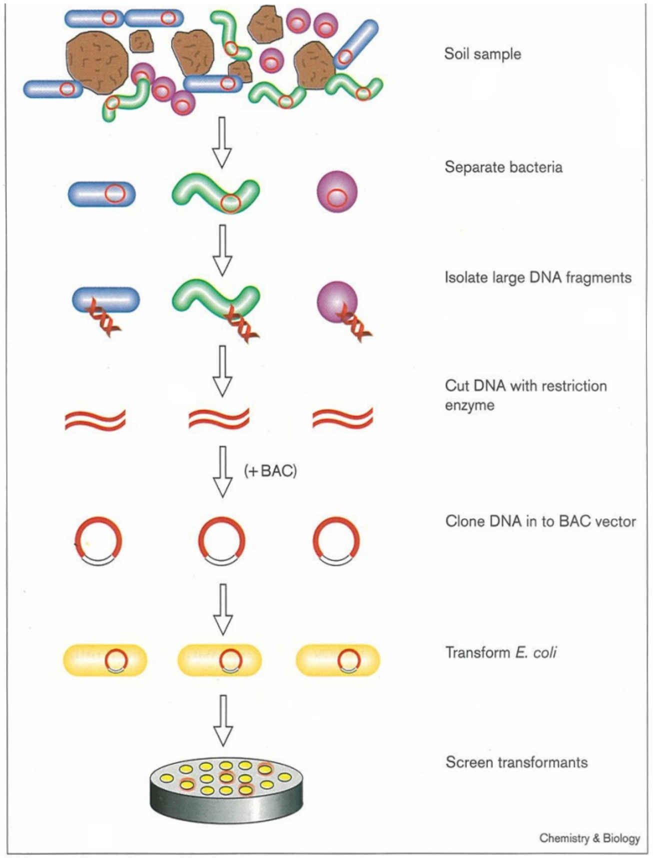


https://doi.org/10.21105/jose.00053

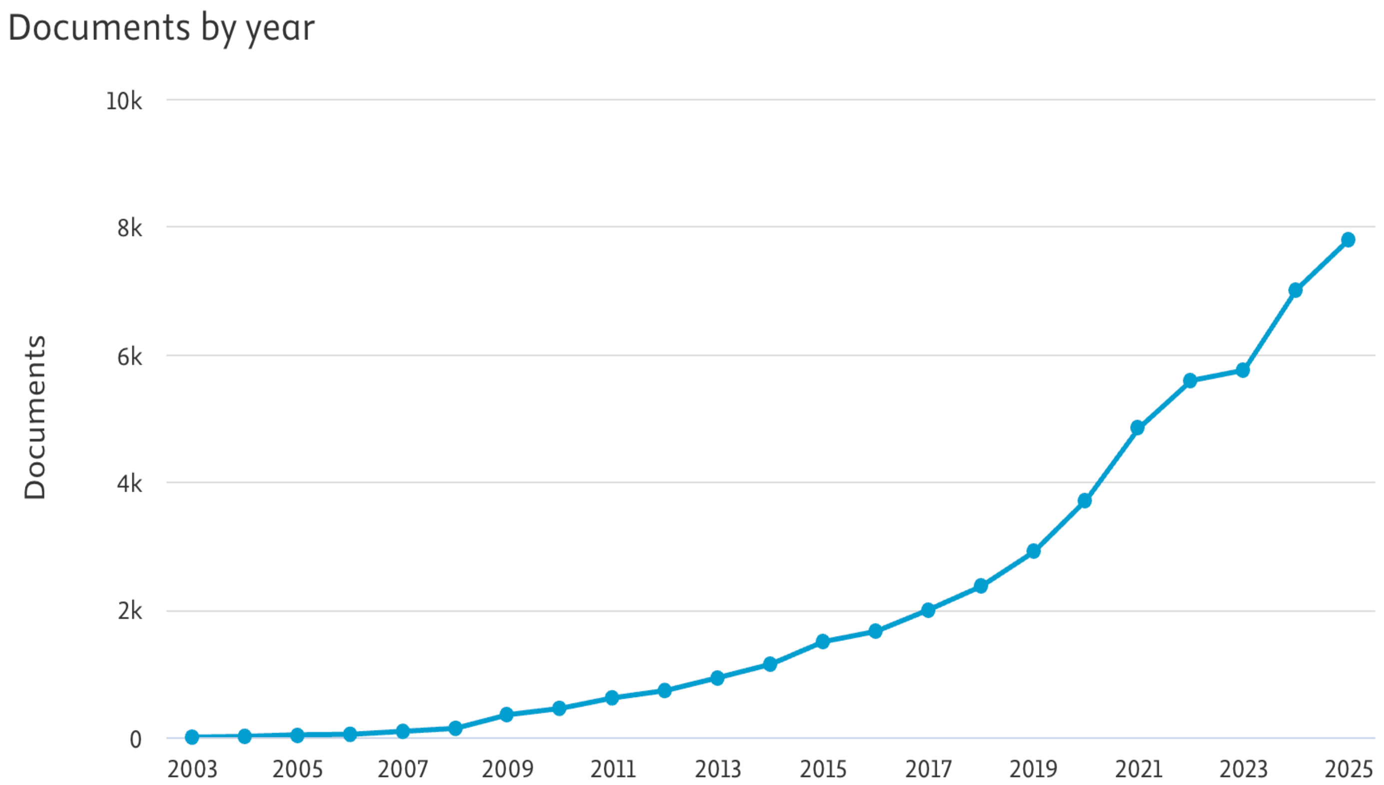
Metagenomics

- Jo Handelsman et al. 1998

Molecular biological access to the chemistry of unknown soil microbes: a new frontier for natural products
Jo Handelsman[1], Michelle R Rondon[1], Sean F Brady[2], Jon Clardy[2] and Robert M Goodman[1]



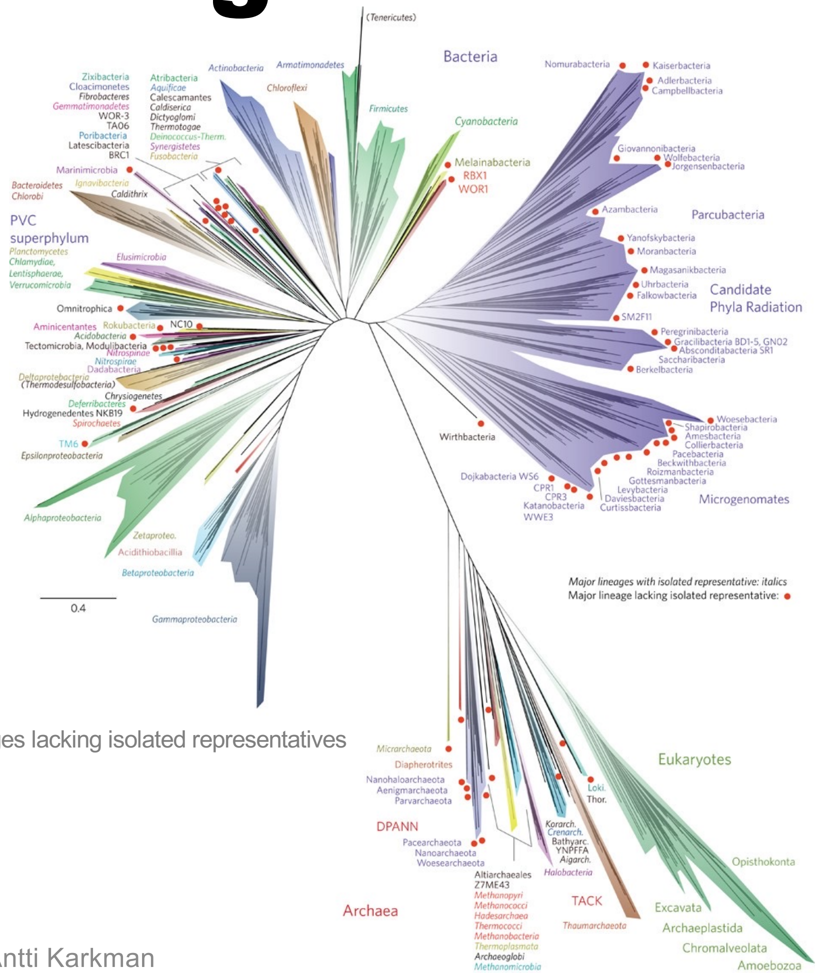
Articles published in metagenomics



Source: Scopus

Why do we need metagenomics?

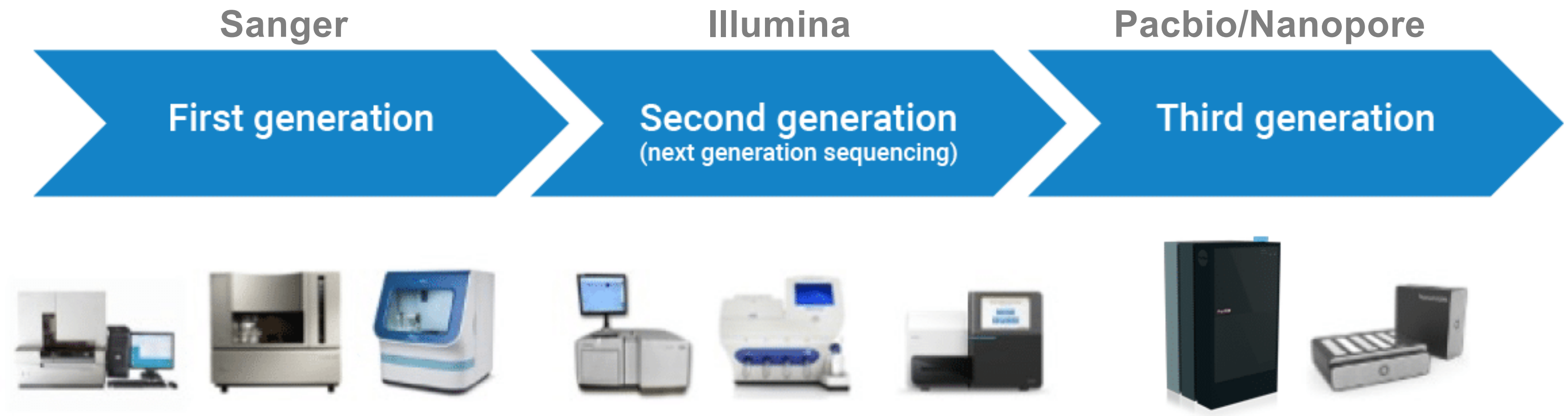
- The great plate count anomaly
- Taxonomy ≠ function



● Lineages lacking isolated representatives

https://doi.org/10.1038/nmicrobiol.2016.48

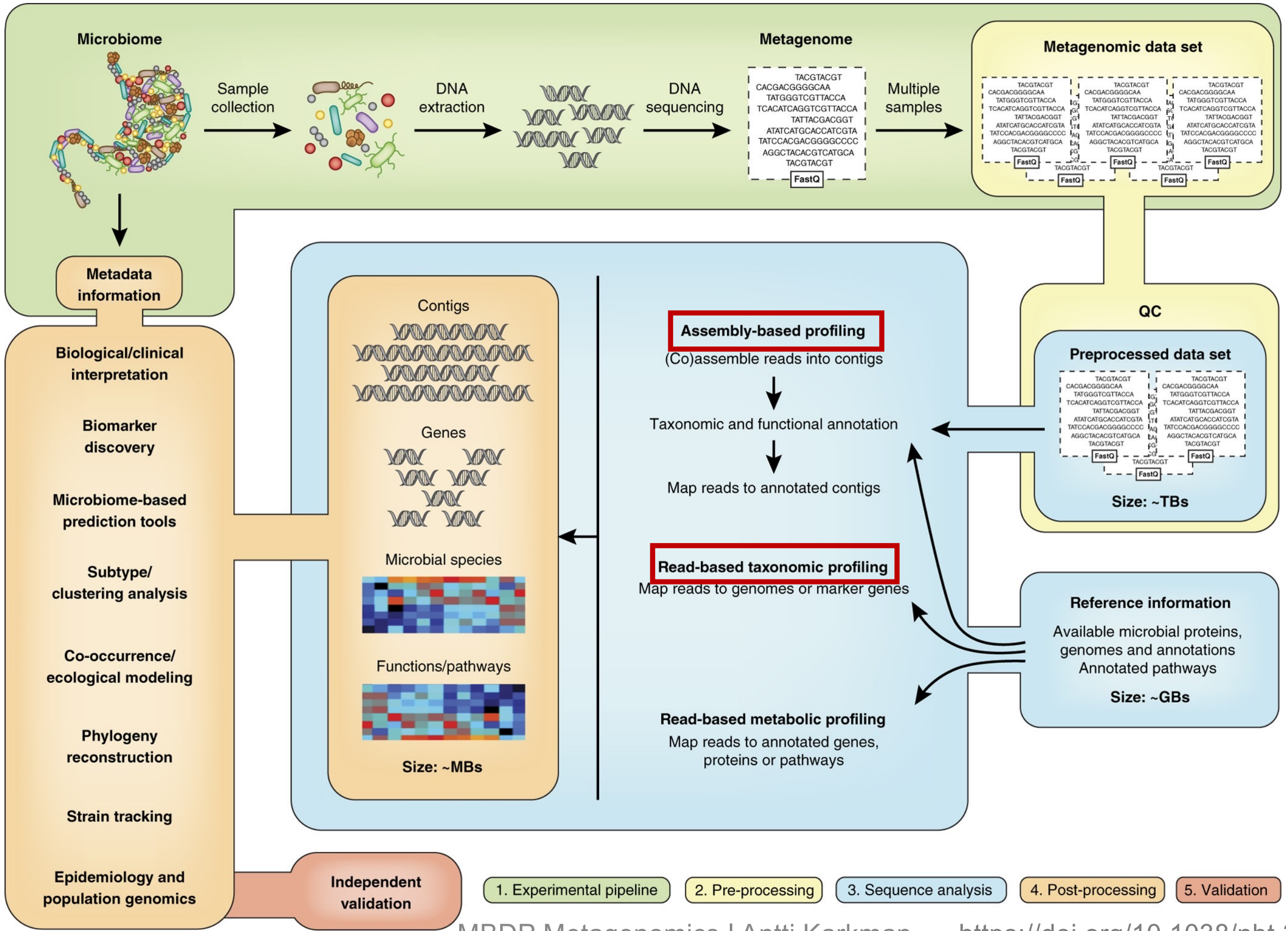
Metagenomic sequencing



Group work:

- 1-3 specific features for each technology
- Which one(s) is/are suitable for metagenomics

Metagenomic data analysis

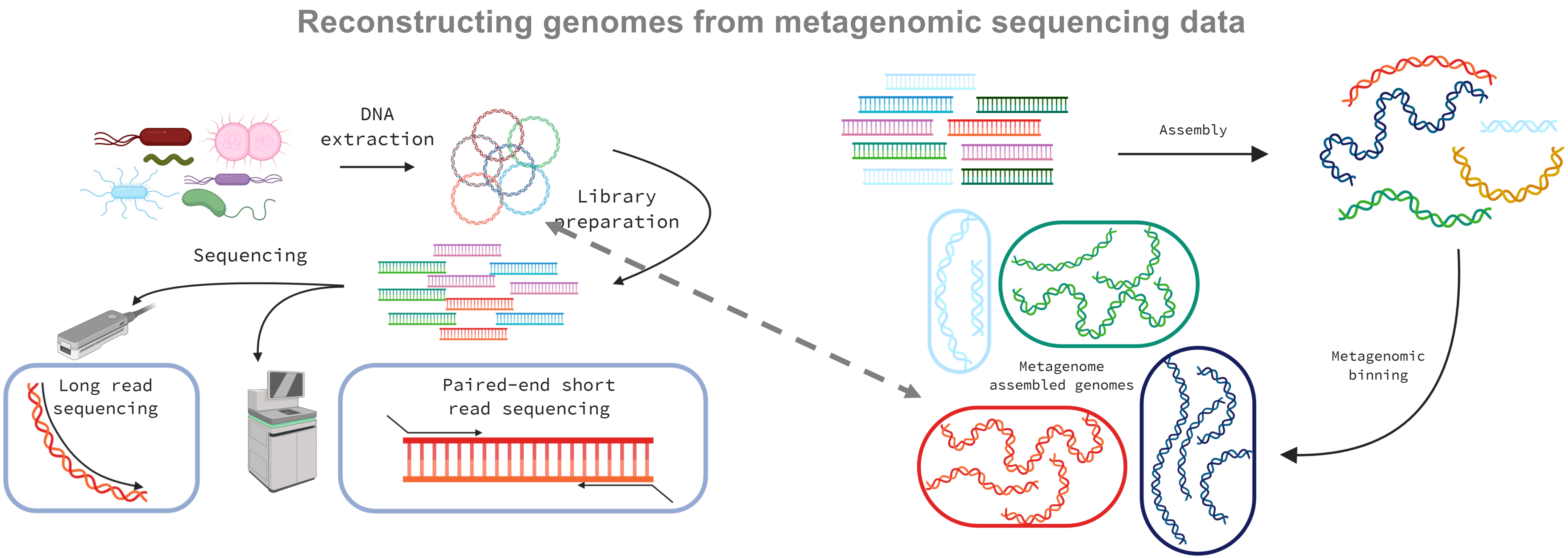


Read-based vs. Assembly-based

Read-based

Assembly-based

Genome-resolved metagenomics



Databases for (meta)genomics

Sequence Read Archive (SRA) European Nucleotice Archive (ENA)

- Publicly available repositories for high-throughput sequencing data
- All sequencing data should be deposited to repositories upon publication
- Sequencing data (Runs) organised under projects (BioProject) and linked to samples (BioSample)
- Various levels of metadata depending on the project
- Web and (several) command line access options

https://www.ncbi.nlm.nih.gov/sra https://www.ebi.ac.uk/ena

MGnify

- Website to browse, analyse, discover and compare microbiome data
- Data from ENA/SRA
- Includes analyses, assemblies and MAG collections

https://www.ebi.ac.uk/metagenomics

Branchwater & Sandpiper

- Annotation of sequencing experiments in SRA/ENA
- Web interface to search for sequences/taxonomy

https://sandpiper.qut.edu.au/ https://branchwater.sourmash.bio/

Time for practicals